4. Mathematical Preliminaries

For some transformations of the AES algorithms specified in Sec. 5, each byte in the state array is interpreted as one of the 256 elements of a finite field, also known as a Galois Field, denoted by GF(2^8). [1]

In order to define addition and multiplication in GF(2^8), each byte $\{b_7\ b_6\ b_5\ b_4\ b_3\ b_2\ b_1\ b_0\}$ is interpreted as a polynomial, denoted by $b(x)$, as follows:

$$b(x) = b_7x^7 + b_6x^6 + b_5x^5 + b_4x^4 + b_3x^3 + b_2x^2 + b_1x + b_0. \tag{4.1}$$

For example, $\{01100011\}$ is represented by the polynomial $x^6 + x^5 + x + 1$.

4.1 Addition in GF(2^8)

In order to add two elements in the finite field GF(2^8), the coefficients of the polynomials that represent the elements are added modulo 2 (i.e., with the exclusive-OR operation, denoted by $\oplus$), so that $1 \oplus 1 = 0$, $1 \oplus 0 = 1$, and $0 \oplus 0 = 0$.

Equivalently, two bytes can be added by applying the exclusive-OR operation to each pair of corresponding bits in the bytes. Thus, the sum of $\{a_7\ a_6\ a_5\ a_4\ a_3\ a_2\ a_1\ a_0\}$ and $\{b_7\ b_6\ b_5\ b_4\ b_3\ b_2\ b_1\ b_0\}$ is $\{a_7 \oplus b_7\ \ a_6 \oplus b_6\ \ a_5 \oplus b_5\ \ a_4 \oplus b_4\ \ a_3 \oplus b_3\ \ a_2 \oplus b_2\ \ a_1 \oplus b_1\ \ a_0 \oplus b_0\}$. (In Section 5.1.4, this definition is extended to words.)

For example, the following three representations of addition are equivalent:

$$\begin{aligned} &(x^6 + x^4 + x^2 + x + 1) + (x^7 + x + 1) = x^7 + x^6 + x^4 + x^2 && \text{(polynomial)} \\ &\{01010111\} \oplus \{10000011\} = \{11010100\} && \text{(binary)} \\ &\{57\} \oplus \{83\} = \{d4\} && \text{(hexadecimal).} \end{aligned} \tag{4.2}$$

Because the coefficients of the polynomials are reduced modulo 2, the coefficient 1 is equivalent to the coefficient –1, so addition is equivalent to subtraction. For example, $x^4 + x^2$ represents the same finite field element as $x^4 - x^2$, $-x^4 + x^2$, and $-x^4 - x^2$. Similarly, the sum of any element with itself is the zero element.

4.2 Multiplication in GF(2^8)

The symbol • denotes multiplication in GF$\left(2^8\right)$. Conceptually, this multiplication is defined on two bytes in two steps: 1) the two polynomials that represent the bytes are multiplied as polynomials, and 2) the resulting polynomial is reduced modulo the following fixed polynomial:

$$m(x) = x^8 + x^4 + x^3 + x + 1. \tag{4.3}$$

Within both steps, the individual coefficients of the polynomials are reduced modulo 2.

[1] Information about the properties of finite fields can be found in textbooks, such as Michael Artin's *Algebra* [4].

Table 4. SBOX(): substitution values for the byte xy (in hexadecimal format)

		y														
		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
x	0	63	7c	77	7b	f2	6b	6f	c5	30	01	67	2b	fe	d7	ab	76
	1	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	c0
	2	b7	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
	3	04	c7	23	c3	18	96	05	9a	07	12	80	e2	eb	27	b2	75
	4	09	83	2c	1a	1b	6e	5a	a0	52	3b	d6	b3	29	e3	2f	84
	5	53	d1	00	ed	20	fc	b1	5b	6a	cb	be	39	4a	4c	58	cf
	6	d0	ef	aa	fb	43	4d	33	85	45	f9	02	7f	50	3c	9f	a8
	7	51	a3	40	8f	92	9d	38	f5	bc	b6	da	21	10	ff	f3	d2
	8	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
	9	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
	a	e0	32	3a	0a	49	06	24	5c	c2	d3	ac	62	91	95	e4	79
	b	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	08
	c	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
	d	70	3e	b5	66	48	03	f6	0e	61	35	57	b9	86	c1	1d	9e
	e	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
	f	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

the substitution value would be determined by the intersection of the row with index '5' and the column with index '3' in Table 4, so that $s'_{r,c} = \{\texttt{ed}\}$.

5.1.2 SHIFTROWS()

SHIFTROWS() is a transformation of the state in which the bytes in the last three rows of the state are cyclically shifted. The number of positions by which the bytes are shifted depends on the row index r, as follows:

$$s'_{r,c} = s_{r,(c+r) \bmod 4} \qquad \text{for } 0 \le r < 4 \text{ and } 0 \le c < 4. \tag{5.5}$$

SHIFTROWS() is illustrated in Figure 3. In that representation of the state, the effect is to move each byte by r positions to the left in the row, cycling the left-most r bytes around to the right end of the row. The first row, where $r = 0$, is unchanged.